Architectural Limitations of Vision–Language Models in Implicit Visual Grounding: Mechanistic Analysis and Diagnostic Experiments

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Abstract

Implicit visual grounding (IVG) requires vision–language models (VLMs) to map natural-language queries with high-level semantics and subjective judgments to precise spatial regions, making it a useful paradigm for analysing spatial–semantic alignment in VLMs. This work proposes a mechanistic framework to characterise the architectural limitations of contemporary VLMs on implicit grounding from five perspectives: visual token compression, the two-dimensional constraints of causal attention, the one-dimensional bias of RoPE positional encoding, gradient conflicts between language modelling and spatial regression objectives, and the progressive loss of fine-grained perception across layers. The framework is centred on formal analysis, including theoretical derivations and upper-bound estimates for each mechanism, as well as their cascading interactions. To provide diagnostic evidence, this study conducts multi-condition information-manipulation experiments on UAV imagery using the Qwen3-VL series. Results show that background suppression consistently improves localisation accuracy, while explicit coordinate injection degrades performance; moreover, Instruct mode consistently outperforms Thinking mode. These observations are consistent with the predictions of the framework. This work aims to provide theoretical tools for understanding and improving VLM architectures and training objectives for implicit visual grounding.

1 Introduction

Vision–language models (VLMs) have achieved substantial progress on benchmarks for visual question answering and multimodal reasoning. Yet these models exhibit a systematic capability gap on a seemingly fundamental task: implicit visual grounding (IVG). Unlike explicit grounding, in which queries such as "locate the person riding a bicycle" refer to directly observable entities, implicit grounding requires the model to map highly abstract semantic concepts onto concrete spatial regions—a mapping that demands the coordinated integration of scene understanding, physical commonsense, and fine-grained spatial localisation. Far from being incidental, this failure pattern systematically exposes structural tensions within the architectural assumptions and training paradigms shared by contemporary VLMs. This work therefore argues that implicit grounding should not be treated as a standalone detection task to be solved, but rather as a diagnostic instrument for evaluating the quality of spatial–semantic alignment—one that reveals specific failure points in the fusion of geometric and semantic information more effectively than existing probing methods or representational analyses.

VLMs are not designed as specialised object-detection tools. Conventional detectors offer rapid inference, low computational overhead, and native batch processing at scale—capabilities that current VLMs cannot match. The value of applying VLMs to implicit grounding lies instead in diagnosis: it serves as a litmus test for whether a model genuinely integrates visual perception, linguistic understanding, and commonsense reasoning. Consistent with this framing, experiments show that even state-of-the-art VLMs, including GPT-4V, Qwen3-VL, and Claude-3.5-Sonnet, achieve accuracy on implicit grounding tasks that falls well below that of dedicated detection models.

Existing work on visual grounding can be broadly divided into two streams. The first comprises dedicated detection architectures, exemplified by SAM, which localise target regions through text-based queries. These approaches perform strongly on explicit grounding tasks but depend on predefined category taxonomies or segmentation prompts and cannot accommodate open-world implicit semantics. The second stream adapts VLMs to produce bounding-box coordinates in text form. Performance in implicit grounding scenarios, however, remains unsatisfactory. Several explanations have been proposed for the broader limitations of VLMs on fine-grained visual tasks, including insufficient visual token budgets, the absence of dedicated spatial-perception training, restricted spatial modelling capacity of causal attention, and limited expressiveness of positional encodings. Each of these factors constitutes a necessary but not sufficient condition for the observed failures. Critically, the factors interact: token compression incurs upstream information loss that no downstream attention mechanism can recover; spatial modelling deficits in the attention layers further degrade the exploitation of whatever spatial information remains; and the language-modelling training objective steers optimisation towards semantic comprehension at the expense of spatial precision. To these known factors, this work adds the progressive attenuation of fine-grained visual features across successive network layers, thereby establishing a comprehensive diagnostic framework for VLM limitations on implicit visual grounding.

2 Implicit Visual Grounding: Task Formulation and Complexity

2.1 Formal Task Definition

Given an input image $I \in \mathbb{R}^{H \times W \times 3}$ and a natural language query Q, the objective of implicit visual grounding is to predict a spatial bounding box $B = (x_1, y_1, x_2, y_2)$ whose enclosed region best matches the implicit semantic concept expressed by Q.

Formally, let $\phi_v(I) \in \mathbb{R}^{N \times d_v}$ denote the image feature representation extracted by a visual encoder, where N is the number of visual tokens and d_v is the feature dimension; and let $\phi_l(Q) \in \mathbb{R}^{M \times d_l}$ denote the query semantic representation produced by a language encoder, where M is the number of query tokens and d_l is the language embedding dimension. Implicit grounding seeks to learn a mapping f such that

$$f\big(\phi_v(I),\ \phi_l(Q)\big) \ \rightarrow\ B = (x_1,\ y_1,\ x_2,\ y_2), \tag{1}$$

where (x_1, y_1) and (x_2, y_2) are the top-left and bottom-right corners of the bounding box, respectively, normalised to $[0, 1000]$. Localisation quality is evaluated by the Intersection over Union (IoU):

$$\mathrm{IoU}(B,\ B_{\mathrm{gt}}) \ = \ \frac{|B \ \cap \ B_{\mathrm{gt}}|}{|B \ \cup \ B_{\mathrm{gt}}|}. \tag{2}$$

A localisation is deemed successful when $\mathrm{IoU}(B, B_{\mathrm{gt}}) \ \geq \ \tau$, where $\tau = 0.5$ serves as the conventional threshold for *acceptable* localisation and $\tau = 0.75$ for *precise* localisation.

2.2 Task Complexity: Implicit versus Explicit Grounding

Semantic complexity. In explicit grounding, the semantic content of query Q maps directly onto observable visual attributes of the target region: colour, shape, and spatial relations can all be resolved through single-hop reasoning over the visual space. In implicit grounding, by contrast, Q encodes compound semantics in which individual sub-concepts—for example, "violating traffic rules"—require additional commonsense reasoning before they can be converted into observable visual cues such as riding against the traffic flow or occupying a motor-vehicle lane.

Reasoning chain length. The reasoning chain in explicit grounding has a length of one: a single mapping step from attribute matching to region selection. In implicit grounding, the chain has a length of at least three: (i) multi-hop reasoning from an abstract concept to relevant commonsense knowledge; (ii) conceptual mapping from that knowledge to concrete visual cues; and (iii) spatial reasoning from visual cues to a target region.

Information requirements. Explicit grounding requires only the target's appearance features and spatial reference, both extractable directly from the image. Implicit grounding additionally demands world knowledge, physical commonsense, and social norms—information that cannot be derived from a single image and must instead be drawn from the model's pre-trained parametric knowledge or from external knowledge bases.

Localisation precision. Implicit grounding imposes stricter precision demands. Mis-localising an agent can yield entirely incorrect conclusions—for instance, classifying a rule-abiding cyclist as a traffic violator. The target agent may, moreover, be small, partially occluded, or spatially entangled with visually similar objects in complex scenes.

3 Five Core Limitations of Implicit Visual Grounding

3.1 Irreversible Spatial Information Loss from Visual Token Compression

The first critical information bottleneck in VLM architectures arises at the ViT-to-MLP fusion stage.

Consider a standard configuration: an input image of 224×224 is divided into patches of 16×16 pixels, yielding a $14 \times 14 = 196$ grid. After passing through the ViT encoder's embedding layer, these 196 patches produce 196 visual tokens of dimension d_v, denoted $T_v \in \mathbb{R}^{196 \times d_v}$. In early VLMs such as LLaVA, these tokens are projected directly to dimension d_l via an MLP and fed into the LLM. In modern architectures such as Qwen3-VL, however, an additional Patch Merger applies 2×2 spatial compression:

$$T_v' = \mathrm{PatchMerger}(T_v), \quad T_v' \in \mathbb{R}^{49 \times d_l}. \tag{3}$$

After this step, the spatial extent represented by each token expands from 16×16 to 32×32 pixels, reducing spatial resolution by a factor of four. From an information-theoretic perspective, the mutual information loss introduced by this compression can be formalised as follows.

Let $H(I)$ denote the pixel-level information entropy of the original image, and $H(\mathrm{patch}_i)$ the information content of each patch i. At the patch-partitioning stage, the total information content is

$$H(P) = \sum_{i=1}^{196} H(\mathrm{patch}_i) \; - \; I(\mathrm{patch}_1, \dots, \mathrm{patch}_{196}), \tag{4}$$

where the second term captures the mutual information among patches arising from the correlation structure learned by ViT self-attention. After the ViT encoder, the conditional mutual information of each token t_i satisfies

$$I(t_i;\, \mathrm{patch}_i) \;\leq\; H(\mathrm{patch}_i), \tag{5}$$

indicating that the ViT representation cannot increase the information content of any individual patch. After the Patch Merger, each compressed token c_j aggregates information from four original patches:

$$I\big(c_j;\, \mathrm{patch}_{4j-3}, \dots, \mathrm{patch}_{4j}\big) \;\leq\; \min_i \; I(t_i;\, \mathrm{patch}_i), \quad i \in \{4j-3, \dots, 4j\}. \tag{6}$$

Because the compression mapping is lossy and information cannot be created, the mutual information that c_j retains about any individual patch is strictly less than that carried by the corresponding uncompressed token t_i. This loss is irreversible: no decoder can recover precise per-patch information from c_j alone.

For implicit visual grounding, spatial details often constitute the critical localisation cues. Detecting a traffic-rule violation, for example, may hinge on distinguishing the riding direction relative to lane orientation—requiring precise spatial relations between the bicycle heading and the lane axis—or on identifying the boundary between motor-vehicle and non-motor-vehicle lanes, or on counting the number of riders and their relative positions to assess overloading. All of these are fine-grained behavioural features. Once discarded during token compression, even a model with powerful language-reasoning capabilities lacks sufficiently detailed spatial information in its visual input to support precise localisation.

The DeepStack mechanism introduced in Qwen3-VL injects intermediate ViT features into the early layers of the LLM, providing richer multi-level visual information. However, because DeepStack adds these features to the first few LLM layers only, the injected representations still undergo progressive semantic abstraction in all subsequent layers. DeepStack increases the hierarchical diversity of the input signal but does not alter the fundamental trend by which fine-grained spatial details are progressively attenuated as information flows through the LLM.

3.2 Theoretical Limits of Causal Attention in Spatial Relationship Modelling

The LLM backbone of mainstream VLMs universally adopts autoregressive (causal) self-attention. Although this design aligns with the unidirectional generation logic of text, it introduces structural representational deficiencies when applied to the unordered, bidirectionally interconnected nature of 2D visual scenes.

Causal masking severs bidirectional spatial links. Consider two spatially adjacent visual tokens t_i and t_j ($i < j$). In 2D physical space, these tokens share a symmetric, bidirectional association. Under the LLM's causal lower-triangular mask, however, only the later token t_j may attend to the earlier token t_i; the reverse attention is forcibly zeroed ($A_{i,j} = 0$). This mechanism produces a severe information deficit for tokens at the front of the sequence: during multi-layer Transformer forward propagation, front-positioned visual tokens cannot access information from regions behind them through any pathway. Local visual details encoded by an early token t_1 cannot be semantically calibrated through bidirectional interaction with downstream global context encoded by a later token t_2, resulting in substantial representational distortion.

One-dimensional flattening destroys 2D spatial structure. Image patches are forcibly flattened into a one-dimensional sequence before entering the LLM, distorting native 2D spatial proximity into 1D sequence proximity and eliminating the geometric translation invariance inherent in two-dimensional layouts. Combined with the causal mask, this flattening imposes a virtual temporal axis onto physical space, converting the bidirectional free flow of spatial information into a unidirectional, lossy chain. This explains a key observation: although deep VLMs can aggregate global semantics through rear-positioned text tokens to form holistic image abstractions, the broken spatial dependency chains among front-positioned visual tokens prevent the model from reconstructing precise spatial geometric coordinates at the representation level—a limitation that manifests acutely in dense prediction tasks such as implicit grounding.

3.3 One-Dimensional Limitations of RoPE Positional Encoding

In an ideal visual attention mechanism, spatially adjacent visual tokens (such as t_{k-1} and t_k) should maintain high mutual attention weights to preserve local geometric continuity. In VLMs built on LLM backbones, however, this spatial proximity is severely undermined at both the representational and structural levels.

Representational level: 1D RoPE spatial distortion and semantic domination of the dot product. In the absence of explicit 2D structural constraints, the LLM relies on Rotary Position Embedding (RoPE) to encode positional information. For a 1D-flattened visual sequence, the attention score between query q_k at position k and key k_{k-1} at position $k-1$, after RoPE transformation, is

$$\left\langle q_k^{(\mathrm{rot})},\ k_{k-1}^{(\mathrm{rot})} \right\rangle \ = \ \left\langle q_k,\ R(1)\, k_{k-1} \right\rangle, \tag{7}$$

where $R(1) = R^{\top}(k)\, R(k-1)$ encodes a relative positional bias for a step size of one. This mechanism suffers from two critical flaws.

Vertical geometric distortion. Because the LLM employs one-dimensional RoPE, two tokens that are vertically adjacent in the 2D image—at coordinates (i, j) and $(i+1, j)$—receive an index separation of W (the image width) in the flattened sequence. RoPE therefore assigns them the relative position matrix $R(W)$. Owing to the long-range decay property of RoPE, the model interprets these spatially neighbouring tokens as extremely distant, effectively severing local geometric associations along the vertical axis.

Semantic domination of the dot product. Even for tokens separated by a horizontal distance of one, the base attention score is governed by the raw dot product $\langle q_k, k_{k-1} \rangle$. If adjacent tokens belong to different semantic categories—for example, the boundary of a cyclist versus the background road surface—their representational dot product approaches zero. The RoPE rotation can only fine-tune attention weights among semantically similar tokens; it cannot reverse the attention collapse caused by semantic dissimilarity.

Structural level: autoregressive Softmax competition and the causal upper bound. Because of the LLM's autoregressive nature, Softmax normalisation operates over the full global context, encompassing both historical text tokens and previously input visual tokens.

Let t_k denote the current query visual token and t_{k-1} its spatially adjacent predecessor. Under causal self-attention, the attention weight $\alpha_{k,k-1}$ that t_k assigns to t_{k-1} satisfies the following causal upper bound:

$$\alpha_{k,k-1} \ \leq \ \frac{\exp\!\left(\dfrac{\langle q_k^{(\mathrm{rot})},\ k_{k-1}^{(\mathrm{rot})} \rangle}{\sqrt{d}} \right)}{\exp\!\left(\dfrac{\langle q_k^{(\mathrm{rot})},\ k_{k-1}^{(\mathrm{rot})} \rangle}{\sqrt{d}} \right) \ + \ \sum_{m \in \mathcal{T}} \exp\!\left(\dfrac{\langle q_k^{(\mathrm{rot})},\ k_m^{(\mathrm{rot})} \rangle}{\sqrt{d}} \right)}, \tag{8}$$

where $\mathcal{T}$ denotes the set of system-prompt and historical text-context tokens preceding t_k in the sequence.

This upper bound reveals that the attention share allocated to a spatially adjacent token is directly governed by competition from all preceding text tokens in the denominator. In practical multimodal tasks, task instructions—such as "locate the cyclist"—exert powerful semantic gravitational pull, causing the $\sum_{m \in \mathcal{T}} \exp(\cdot)$ term to expand dramatically. This constitutes a mathematical proof that local spatial proximity receives no inherent priority in LLM self-attention: its weight is relentlessly diluted in competition with global text semantics.

3.4 Fundamental Contradiction Between Language Modelling and Spatial Regression Objectives

This section addresses what is arguably the most fundamental limitation of VLMs on implicit visual grounding: a structural misalignment of training objectives.

The VLM training objective is to maximise the likelihood of a conditional language model:

$$\mathcal{L} = -\sum_{t=1}^{T} \log P(w_t \mid w_{<t},\, I;\, \theta), \tag{9}$$

where w_t is the t-th text token, I is the input image, and T is the sequence length. In essence, this objective trains the model to generate plausible text sequences conditioned on image context. What it optimises is fluency, semantic coherence, and task completion in the generated text—not spatial localisation precision.

When a VLM is asked to output bounding-box coordinates, it must express them as a text token sequence—for example, "The person is at [245, 378, 523, 892]." This translation from spatial output to language output introduces four fundamental contradictions.

Contradiction 1: discrete token output versus continuous spatial regression. Bounding-box coordinates $B = (x_1, y_1, x_2, y_2)$ are continuous values with, in principle, infinitely many possible settings. In a language model, however, each coordinate must be encoded as discrete tokens (digits 0–9 and separators). For normalised coordinates in $[0, 1000]$, each axis admits 1,001 possible integer values, each mapped to a distinct vocabulary token. The model produces a probability distribution over discrete character tokens at each decoding step and selects the highest-probability token as output.

Contradiction 2: probability peaks versus precise numerical values. For each coordinate token, the model outputs a conditional probability distribution $P(\text{token} \mid \text{previous tokens}, I)$. Even when the model "knows" the exact position, it must express it through discrete token sequences. The Softmax probability peak may be broad, leaving the model unable to discriminate between nearby coordinate values: "245" and "246" are entirely distinct tokens in the vocabulary, yet their corresponding bounding boxes differ by only a single pixel in image space.

Contradiction 3: absence of spatial gradient supervision. Standard detection or segmentation models are trained end-to-end with objectives that include a direct bounding-box regression term:

$$\mathcal{L}_{\text{detect}} = \mathcal{L}_{\text{cls}} + \lambda \cdot \mathcal{L}_{\text{box}}, \quad \mathcal{L}_{\text{box}} = \sum_i \text{SmoothL1}(b_i,\ b_i^{\text{gt}}), \tag{10}$$

where $\mathcal{L}_{\text{box}}$ provides gradients that flow directly from the output layer back to spatially aware feature representations. In a VLM, by contrast, the language-modelling loss $\mathcal{L}_{\text{LM}}$ supplies only an indirect and heavily diluted gradient signal for localisation: improvements in localisation accuracy do not translate directly into reduced perplexity, because $\partial\mathcal{L}_{\text{LM}}/\partial B \ll \partial\mathcal{L}_{\text{box}}/\partial\text{backbone}$.

Contradiction 4: bounding boxes as a by-product of language generation. During VLM training, bounding-box coordinates appear within the data annotations, but their accurate prediction is not the primary optimisation target. The language-modelling objective imposes no direct reward or penalty for localisation precision. The model may learn to produce bounding-box tokens because doing so reduces language loss, but no mechanism compels it to output those coordinates accurately.

3.5 Progressive Attenuation of Fine-Grained Visual Perception with Network Depth

3.5.1 Definition of Spatial Heterogeneity Metric

A critical mechanism in multimodal models is that fine-grained spatial details embedded in visual features undergo progressive attenuation and spatial resolution collapse as they propagate through the deeply stacked Transformer layers of the LLM. This phenomenon renders deep representations increasingly inadequate for downstream tasks requiring precise spatial localisation.

Let the LLM comprise L layers, and let the input visual token sequence be $\mathbf{H}^{(0)} = \left[h_1^{(0)}, h_2^{(0)}, \dots, h_N^{(0)}\right]^\top \in \mathbb{R}^{N\times d}$, where N is the number of visual tokens. To quantify the degree to which the l-th layer's representation preserves fine-grained spatial discriminative information, this study defines the Spatial Representation Heterogeneity (SRH):

$$D_{\text{spatial}}\big(\mathbf{H}^{(l)}\big) \;=\; \frac{1}{N(N{-}1)} \sum_{i\neq j} \Big[1 - \cos\big(h_i^{(l)},\, h_j^{(l)}\big)\Big], \tag{11}$$

where $\cos(a, b) = \dfrac{a^\top b}{\|a\|_2\,\|b\|_2}$ denotes the cosine similarity. Higher values of D_{spatial} indicate greater representational discrimination across spatial positions; as the metric approaches zero, spatial distinctiveness vanishes entirely and all positions collapse to a shared semantic centroid.

3.5.2 Progressive Attenuation and Spatial Collapse

In a standard Transformer-based LLM, the spatial heterogeneity of visual tokens satisfies a representation collapse bound as the layer depth l increases. Let $\mathbf{H}^{(l)}$ denote the output representation at layer l. Under the joint action of self-attention and the language-biased feedforward network (FFN), there exists a contraction factor $\gamma \in (0, 1)$ such that

$$D_{\text{spatial}}\big(\mathbf{H}^{(l)}\big) \;\leq\; \gamma^{\,l} \cdot D_{\text{spatial}}\big(\mathbf{H}^{(0)}\big) \;+\; \epsilon, \tag{12}$$

where $\epsilon > 0$ is a residual representational offset determined by textual anchors. As $l \to L$, the feature vectors at distinct spatial positions become mutually similar:

$$\lim_{l\to\infty} \cos\big(h_i^{(l)}, h_j^{(l)}\big) \;=\; 1 - \mathcal{O}(\epsilon) \quad \forall\, i, j. \tag{13}$$

3.5.3 Mathematical Proof and Physical Mechanisms

The collapse dynamics are governed jointly by the low-pass filtering property of self-attention and the semantic-space projection of the FFN.

Mechanism 1: self-attention as spatial diffusion (oversmoothing). The self-attention operation can be written as

$$\mathbf{A}^{(l)} = \text{softmax}\left(\frac{\mathbf{Q}^{(l)}\big(\mathbf{K}^{(l)}\big)^{\top}}{\sqrt{d}}\right), \quad \mathbf{H}^{(l)} = \mathbf{A}^{(l)}\mathbf{V}^{(l)}, \tag{14}$$

where $\mathbf{A}^{(l)} \in \mathbb{R}^{N\times N}$ is a row-stochastic matrix. By Markov chain theory and spectral graph theory, repeated multiplication by a row-stochastic matrix is equivalent to heat diffusion on the token graph. Because the largest eigenvalue of $\mathbf{A}^{(l)}$ is $\lambda_1 = 1$ (corresponding to the constant eigenvector) and the second-largest eigenvalue satisfies $\lambda_2 < 1$, each attention update drives inter-token differences to decay at least as fast as λ_2^l—that is, exponentially. In the framework of graph signal processing, fine-grained spatial details correspond to high-frequency spatial signals. Self-attention therefore functions as an intrinsic low-pass filter: as l increases, high-frequency spatial variation is rapidly suppressed and all visual token representations converge towards a globally shared semantic centroid.

Mechanism 2: FFN manifold contraction within the language attractor domain. The FFN takes the form

$$\text{FFN}(x) = \sigma(x W_1 + b_1)\, W_2 + b_2. \tag{15}$$

Because the LLM parameters (W_1, W_2) are pre-trained on massive text corpora, their row and column spaces are strongly aligned with the linguistic manifold $\mathcal{M}_{\text{text}}$. Any visual input x_v can be orthogonally decomposed into a language-relevant component $x_{\|} \in \mathcal{M}_{\text{text}}$ and a pure visual-detail component $x_{\perp} \in \mathcal{M}_{\text{text}}^{\perp}$. Under the normalising effect of pre-training, the FFN weight matrices act as a contraction mapping on the non-linguistic orthogonal subspace:

$$\big\|\text{FFN}(x_{\perp})\big\|_2 \;\ll\; \|x_{\perp}\|_2. \tag{16}$$

Consequently, at each FFN layer the purely visual local-detail component $x_{\perp}$ is further suppressed while the language-semantic component $x_{\|}$ is amplified. Geometrically, the visual token sequence is irreversibly drawn towards the LLM's semantic attractor domain.

3.5.4 Adverse Impact on Implicit Localisation

Implicit visual grounding depends critically on high-resolution differential features across spatial regions. From the collapse bounds derived above, when the network depth satisfies $l \geq L_0$, the cosine similarity between representations $h_A^{(l)}$ and $h_B^{(l)}$ of spatially non-adjacent regions A and B approaches unity. This means that although the decoder may still identify at the semantic level that "a pedestrian is present," the model has lost the capacity to spatially disambiguate region A from region B in representation space. This result provides a fundamental theoretical explanation for the persistent precision ceiling observed in deep-feature-based localisation.

4 Experiments

4.1 Experimental Design

The central design principle of the experiments is to manipulate the supply of visual information in order to probe VLM localisation behaviour under varying informational conditions. Five attention-guiding methods, each representing a distinct information-supply regime, are devised:

(1) **Baseline (direct query).** The model's raw detection capability, serving as the reference for all comparisons.

(2) **Prompt (coordinate hint).** A positional hint for the target entity is appended to the query, testing the model's ability to exploit positional priors.

(3) **Blur (Gaussian blur).** Regions outside the target area are Gaussian-blurred, preserving global context while suppressing local detail.

(4) **Blackout (region masking).** Regions outside the target area are masked entirely in black, testing the model's capacity for reasoning under severe information deficit.

(5) **Crop (region cropping).** Only local content surrounding the target region is retained, testing the model's reliance on fine-grained local perception.

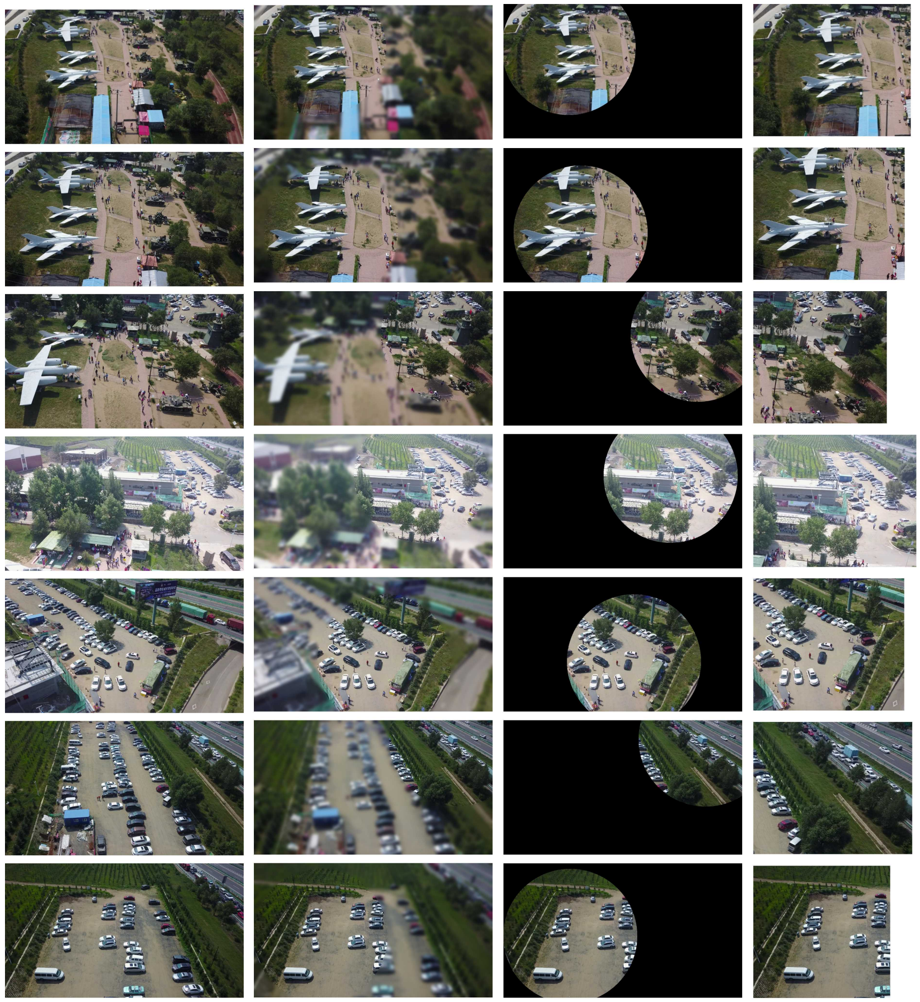

Figure 1: Illustration of the five attention-guiding methods for probing VLM localisation behaviour.

The correspondence between these methods and the theoretical framework is as follows. Baseline directly measures the model's innate visual perceptual capacity. Blur and Blackout manipulate the granularity of available visual information, thereby testing the spatial-detail-loss theory and the progressive-attenuation theory. Prompt injects supplementary positional information to probe the quality of semantic–spatial alignment (Sections 3.3–3.4). Crop strips away global context, isolating the model's dependence on local fine-grained perception and its vulnerability to the loss of long-range spatial dependencies (Section 3.2).

Experiments are conducted on the DVGBench remote-sensing image dataset, which comprises 50 images, each annotated with entity location labels and a natural-language query. Evaluation metrics are Acc@0.5 (classification accuracy at an IoU threshold of 0.5) and average IoU (Avg IoU).

Four model configurations are evaluated: Qwen3-VL-2B-Instruct, Qwen3-VL-4B-Instruct, Qwen3-VL-8B-Instruct, and Qwen3-VL-8B-Thinking.

The test set comprises 50 images. This sample size falls within the small-sample regime and accordingly constrains the nature of the conclusions: many results are presented as trends rather than definitive findings. For cross-method or cross-model comparisons, this study employs basic difference calculations and effect-size descriptions to support the judgements, but do not conduct formal statistical significance tests. Where conclusions rest on trend-level observations, the text explicitly marks them with qualitative qualifiers such as “preliminary evidence suggests” to distinguish them from high-confidence, large-sample findings.

4.2 Experimental Results

(1) **Model-scale effect.** Within the Instruct series, the 8B model achieves the highest or tied-highest Acc@0.5 across all five methods. The 4B model occupies an intermediate position, and the 2B model ranks lowest overall, though not uniformly—its Baseline Acc@0.5 of 0.60 exceeds the 4B model’s 0.58. This pattern of positive but non-monotonic scaling is consistent with the expectation that larger models enhance both semantic comprehension and spatial reasoning, albeit with diminishing marginal returns.

(2) **Universal performance degradation under Prompt.** Across all four model configurations, the Prompt method consistently degrades performance, with no exceptions. The most extreme case is Qwen3-VL-8B-Thinking, whose Prompt accuracy collapses from a Baseline of 0.50 to 0.08—a relative drop of 84%.

(3) **Consistent positive effects of Blur and Blackout.** Across all Instruct models, both Blur and Blackout consistently improve performance relative to the Baseline.

(4) **Comprehensive advantage of Instruct mode over Thinking mode.** Instruct mode outperforms Thinking mode across all five methods. The mean Acc@0.5 values are 0.62 and 0.44 respectively, corresponding to an 18 percentage-point gap.

Figure 2: Predicted and ground-truth bounding boxes across the five attention-guiding methods.

4.3 Analysis of Experimental Results

Model scale has limited impact on visual processing but amplifies alignment fragility. Under Baseline, Blur, Blackout, and Crop, the performance of the Qwen3-VL series varies little with parameter count, with overall fluctuations contained within 10%. This suggests that low-level visual feature extraction has largely converged even in lightweight models. Under the Prompt condition, however, the influence of model scale is dramatically amplified: the 2B model suffers an accuracy drop of approximately 30% relative to the 8B model. This extreme vulnerability points to insufficient language–vision alignment stability at small parameter counts. Explicit coordinate prompts impose stringent demands on cross-modal high-dimensional representation mapping and reasoning, and low-parameter models are prone to attention dispersion triggered by context overload.

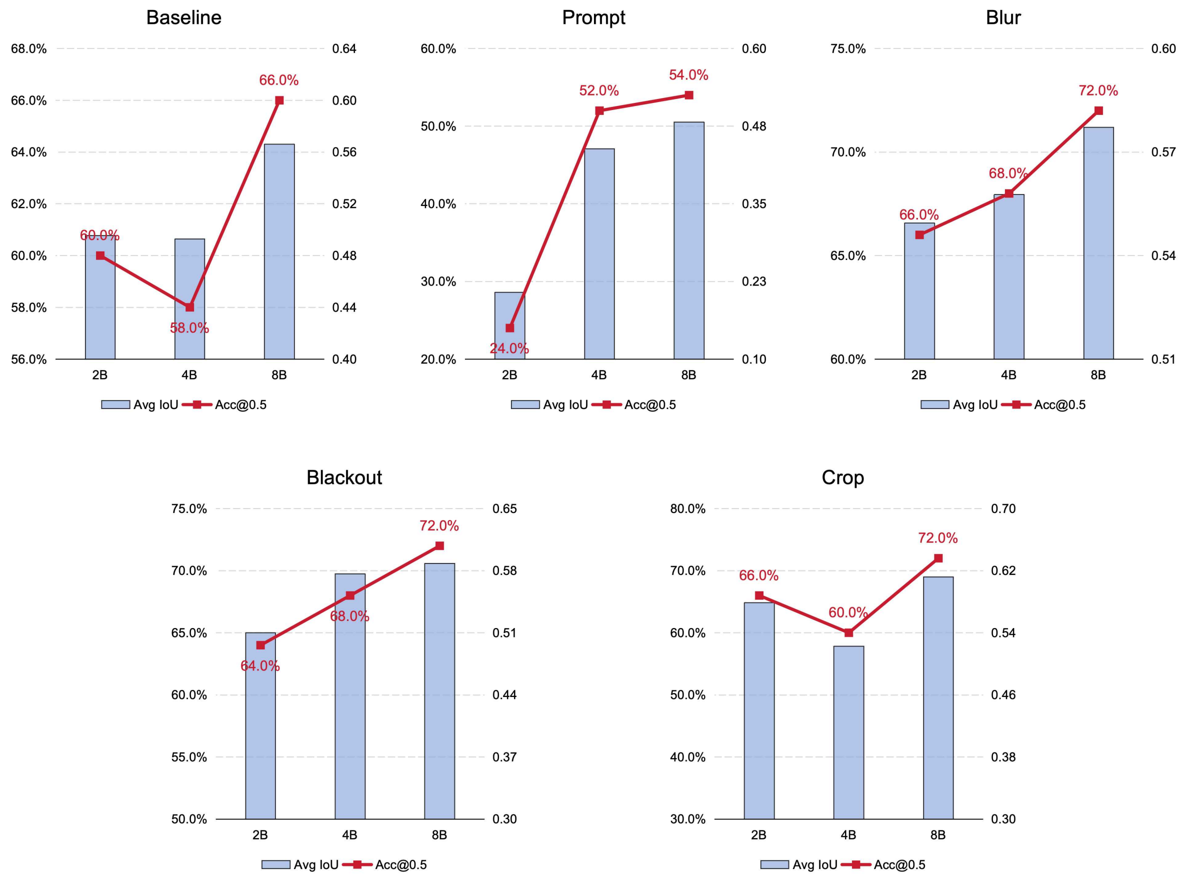

Figure 3: Cross-scale comparison of localisation performance under the five attention-guiding methods.

The universal failure of Prompt reflects structural constraints of language modelling on localisation. This counter-intuitive finding reveals that when coordinate information is injected as token sequences, the language-modelling objective dominates the entire processing pipeline. The model prioritises how coordinates can be naturally expressed and continued in text, rather than the spatial regions those coordinates designate in the image. Moreover, the addition of Prompt tokens forces the model to simultaneously align Prompt tokens, visual tokens, and question tokens, substantially raising alignment difficulty. Achieving coherent three-way alignment demands greater model capacity, which explains the pronounced performance degradation in the 2B configuration.

Information absence paradoxically improves performance. The core commonality shared by Blur, Blackout, and Crop is the reduction of background interference. From an information-theoretic perspective, under unmodified conditions background clutter competes with target information for the limited pool of visual tokens. When the background is suppressed, the target region's share in the compressed representation increases, yielding higher effective information density. This finding provides empirical support for the visual token compression theory: for implicit visual grounding, what matters is not the total volume of visual information received, but the proportion that is genuinely relevant to target localisation. Crop pushes this logic to its extreme: it not only removes background interference but also, through spatial trimming, ensures that the target region occupies the centre of the visual field. When the target is centrally positioned, the spectral properties of the positional encoding remain relatively stable, enabling the model to more readily establish accurate centre-to-periphery spatial relationships.

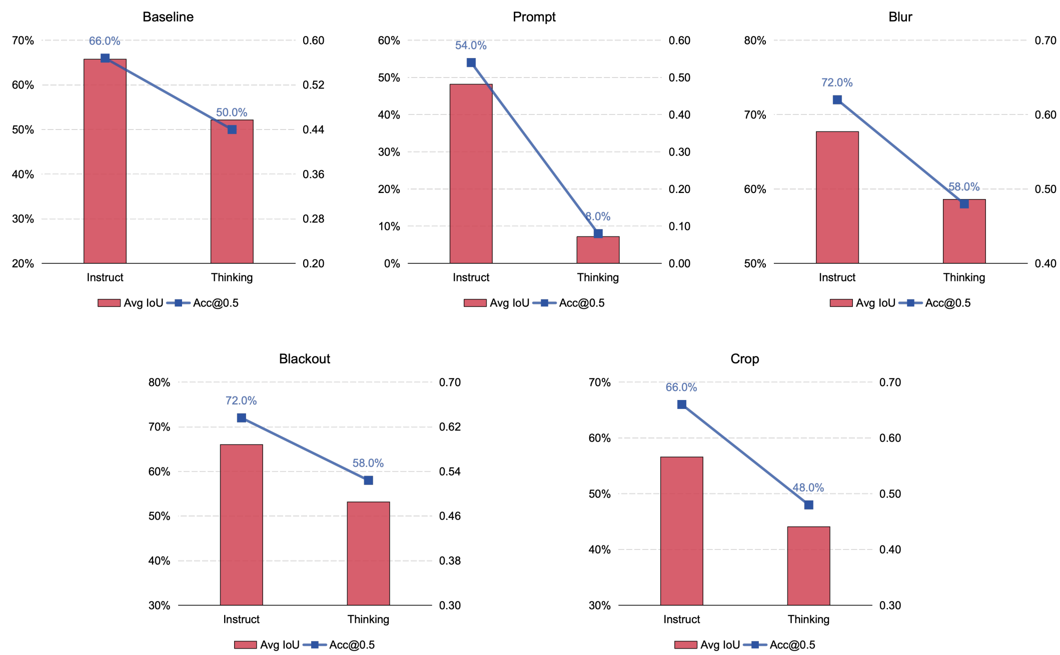

Figure 4: Comparison of Thinking and Instruct modes across the five attention-guiding methods.

Instruct mode outperforms Thinking mode. Thinking mode underperforms Instruct across all five methods, with the gap most pronounced on Prompt. Two factors contribute. First, complex prompts substantially increase the model's intermediate reasoning burden, compressing the available output space for final localisation under a fixed decoding budget. Second, Thinking mode introduces explicit intermediate reasoning steps; from an information-theoretic standpoint, each step constitutes a re-encoding and transmission of information, and each re-encoding risks cumulative error—an effect further amplified when the reasoning chain involves visual information. Notably, under Crop, which deprives the model of global context, Thinking-mode performance degrades slightly more than Instruct-mode performance, indicating that longer reasoning chains create deeper dependence on global reference signals and are therefore more sensitive to contextual removal.

4.4 Mapping Theoretical Limitations to Experimental Evidence

4.4.1 Irreversible Spatial Information Loss from Visual Token Compression

The performance gains observed under Blur, Blackout, and Crop—methods that actively mask background information—carry an important converse implication: under unmodified conditions, the presence of background information exacerbates the negative effects of spatial information compression. Background clutter competes with target information for the limited pool of visual tokens, preventing the spatial details of the target region from being adequately preserved.

4.4.2 Spatial Modelling Limits of Causal Attention and Positional Encoding

Causal attention constrains the efficiency of visual information transfer during autoregressive generation, preventing spatially adjacent information from being preferentially leveraged. When the model's localisation decision relies entirely on the language-modelling pathway represented by the Prompt method, precise positional injection paradoxically becomes a source of interference. This indicates that attention weight allocation among visual tokens is not governed by spatial proximity: the model does not preferentially attend to visual tokens surrounding the target region but instead allocates attention resources to semantically relevant text tokens. The performance gains from Blur and Blackout can thus be interpreted as external interventions that compensate for this intrinsic deficiency of the attention mechanism: when background information is eliminated, the mechanism needs to distribute resources only across a reduced set of visual tokens, circumventing the need for optimal spatial proximity weighting.

Although RoPE formally supports multi-dimensional positional encoding, its effective encoding is constrained by frequency allocation factors, making it difficult for the model to precisely distinguish two-dimensional positional relationships along different axes. The failure of coordinate injection in the Prompt method is linked to this precision limitation. When coordinates are injected as text, the model must interpret the precise semantics of (h, w) two-dimensional coordinates, yet RoPE's spectral interleaving design has inherent limitations when processing exact Cartesian coordinates: the model can comprehend vague spatial descriptions such as "the upper-left region of the image" but struggles to map precise numerical coordinates into corresponding spatial attention distributions. The advantage of Crop over Blur and Blackout on the 8B model (Avg IoU: 0.6119 versus 0.5883 and 0.5772) may partly stem from Crop's forcing of the target region to the visual centre, which simplifies the spatial encoding task: when the target is centrally positioned, the spectral properties of the positional encoding remain relatively stable, enabling the model to more readily establish accurate centre-to-periphery spatial relationships.

4.4.3 Contradiction Between Language Modelling and Spatial Regression

The cross-entropy loss of language modelling provides an extremely weak gradient signal for localisation, and the quantisation errors and error-propagation effects introduced by text decoding further limit localisation precision. This limitation receives the strongest empirical support from the experiments. The failure of the Prompt method fundamentally reveals the structural constraint that the language-modelling objective imposes on localisation: when positional information is injected as natural-language tokens, the language-modelling objective dominates the entire processing pipeline, and the model prioritises optimising "how this coordinate information is naturally expressed and continued" over "which spatial region in the image this coordinate designates." The extreme vulnerability of Thinking mode further corroborates this interpretation: longer reasoning chains amplify the dominance of the language-modelling objective, further degrading localisation performance.

4.4.4 Progressive Attenuation of Fine-Grained Visual Perception

In the LLM backbone of a VLM, fine-grained spatial information in visual tokens diminishes progressively through successive Transformer layers, with deep representations retaining little beyond high-level semantic abstraction. Crop's superior performance on the 8B model (highest Avg IoU of 0.6119) serves as an experimental indicator of this limitation. Crop forces the model to rely on local information for localisation, which better preserves access to fine-grained spatial perception—enabling the model to identify the target's precise edges and local textural features. If the model's deep representations still retained sufficient fine-grained information, Crop should not markedly outperform Baseline. Yet the results show that Crop's Avg IoU (0.6119) exceeds that of Baseline (0.5660), indicating that Crop's local-information focus indeed helps the model better exploit residual fine-grained detail. Conversely, this implies that under Baseline conditions the fine-grained spatial information preserved in the model's deep representations is already insufficient; the model therefore requires Crop's forced attentional concentration to make effective use of whatever residual detail remains.

In the Qwen2.5-VL-7B model used by Zhou et al., the base model achieves an accuracy of 54.75% on the full DVGBench test set, whereas Qwen3-VL-8B reaches 70.41%. Beyond training improvements, this gain is plausibly attributable to the DeepStack mechanism introduced in Qwen3-VL-8B. DeepStack enables shallow-layer surface spatial relationships from the ViT to be transmitted directly to the LLM, circumventing the situation in which visual tokens arriving at the LLM retain only deep, globally abstracted semantic information.

4.4.5 Information Absence as a Positive Effect: An Information-Theoretic Interpretation

From an information-theoretic perspective, the VLM in the experiments can be framed as a reasoning system operating under information-constrained conditions. Under Baseline, the model receives the full visual input, but this input has undergone severe token compression, with substantial fine-grained spatial information lost in the process. The model must therefore perform localisation reasoning with incomplete compressed information, and its accuracy is bounded by this incompleteness.

When Blur or Blackout is applied, background information is suppressed and a higher proportion of the visual tokens entering the model correspond to the target region. Although the total volume of information decreases, the proportion relevant to localisation increases—information quality rises even as information quantity falls. Crop pushes this logic to its extreme: it not only removes background interference but also, through spatial trimming, ensures that the target region occupies the core of the visual focal field.

This interpretive framework aligns closely with the signal-to-noise ratio concept in Shannon information theory: when noise is effectively suppressed, the discernibility of the useful signal improves even if the total signal volume decreases, thereby enhancing reasoning quality. For VLM-based localisation, the critical factor is not how much visual information is received, but how much of it is genuinely devoted to target localisation.

5 Systematic Review and Future Directions of Current Mitigation Strategies

5.1 Taxonomy and Critique of Existing Mitigation Strategies

Although VLMs face fundamental architectural limitations on implicit visual grounding, both academia and industry have explored several mitigation strategies.

Increasing the information density of visual tokens. Representative works include the high-resolution scheme of LLaVA-1.5, the any-resolution technique of InternVL, and the DeepStack cross-layer feature fusion of Qwen3-VL. These strategies share a common principle: increasing the information carried by each token without increasing the total token count. They do improve visual information quality—as noted above, DeepStack delivers substantial gains for the Qwen3-VL series. However, their effectiveness is bounded. DeepStack, for instance, injects intermediate ViT features into the early LLM layers, enabling the LLM to receive both shallow spatial information and high-level global semantic information simultaneously. Yet regardless of how rich the input is, deep-layer processing within the LLM still causes progressive attenuation of fine-grained spatial detail. DeepStack mitigates the problem at the input side, but the structural attenuation at the processing side remains unaddressed.

Improving the spatial modelling capacity of positional encoding. Representative works include the Interleaved MRoPE of Qwen3-VL and the 3D positional encoding of GEMINI. The rationale is to enable positional encodings to better capture spatial structure, thereby directing attention towards spatially adjacent tokens. Interleaved MRoPE improves spectral balance, but the mathematical relationship between attention weights and spatial distance admits an irreducible upper bound: the relative-position term contributed by RoPE is merely one additive component in the attention score, whereas semantic similarity remains the dominant factor in Softmax normalisation. In other words, improved positional encoding can at most help the model locate spatially adjacent tokens more readily; it cannot compel the model to prioritise them.

Hybrid architectures. Representative works include Grounding DINO combined with GPT-4V, Kosmos-G, and Pix2Struct-Enhanced. These approaches first employ a dedicated detector to generate high-quality region proposals, then rely on the VLM to perform semantic comprehension and filtering. This is currently the most effective mitigation strategy, achieving near-specialised-detector performance on explicit grounding tasks. For implicit grounding, however, the core challenge shifts to generating candidate regions that correspond to implicit semantics. Dedicated detectors can only produce proposals for predefined categories and cannot accommodate open-world implicit concepts. For example, no detector is designed to generate proposals for "traffic-rule violations," because this is a behavioural rather than an object category.

Localisation-oriented post-training. After standard supervised fine-tuning (SFT), the model is further fine-tuned on dedicated implicit grounding data with a joint objective combining the language-modelling loss and a localisation IoU loss. This is intuitively the most direct approach but faces severe practical challenges. First, high-quality implicit grounding annotations are extremely scarce—such data are virtually absent from large-scale datasets including LAION and COCO. Second, even with adequate data, the gradient directions of the two loss terms may conflict: the language-modelling loss drives representations towards semantic abstraction, whereas the IoU loss drives them towards spatial refinement, creating a fundamental competition in representation learning.

5.2 Future Research Directions

This work argues that implicit visual grounding should not be treated as a standalone object-detection task but rather as an analytical instrument for diagnosing the spatial representation capabilities of VLMs. Current failures on this task are unlikely to stem from a lack of specialised detection modules; more probably, they reflect insufficient alignment between visual geometric information and linguistic semantics within the shared multimodal representation space. Several structural deficiencies remain prevalent in existing architectures: the destruction of spatial proximity relationships during one-dimensional visual tokenisation, the excessive dilution of low-level geometric detail by high-level semantic features, and the restriction of global spatial interaction by autoregressive causal attention.

Future work should therefore prioritise enhancing the VLM's native spatial modelling and multimodal alignment capabilities while preserving the consistency of its general-purpose generative architecture, rather than introducing additional detection heads or specialised localisation branches. Promising directions include: more flexible multimodal attention mask designs; visual tokenisation methods with greater spatial preservation capacity; lightweight cross-modal projection mechanisms that retain geometric structure; and chain-supervision strategies tailored to spatial reasoning processes. The overarching objective is not to convert the VLM into a dedicated detector but to strengthen its intrinsic spatial understanding and multimodal reasoning capacity without compromising its generality.

6 Conclusion

This work presents a mechanistic framework that identifies five interconnected architectural limitations underlying the systematic failure of vision–language models on implicit visual grounding: visual token compression, the bidirectional-access deficit of causal attention, the one-dimensional bias of RoPE positional encoding, the structural misalignment between language-modelling and spatial regression objectives, and the progressive attenuation of fine-grained perception across network depth. Crucially, these limitations are not independent failure modes but form a cascading chain—token compression causes upstream information loss that no downstream attention mechanism can recover, causal masking prevents spatially coherent re-integration, RoPE further distorts residual spatial relations, the language-modelling training objective steers representations away from geometric precision, and deep-layer processing progressively erodes whatever fine-grained detail survives. Multi-condition information-manipulation experiments on the DVGBench dataset using the Qwen3-VL series provide diagnostic evidence consistent with each predicted mechanism.

Rather than proposing another detection module, this work argues that implicit visual grounding should be repurposed as a diagnostic instrument for probing the quality of spatial–semantic alignment in multimodal models. The five limitations identified here are architectural in nature and therefore generalise beyond any single model family. Addressing them will require coordinated advances in tokenisation, attention design, positional encoding, and training objectives—advances that strengthen the VLM's intrinsic spatial understanding without sacrificing its general-purpose generative capability.